

Machine Learning

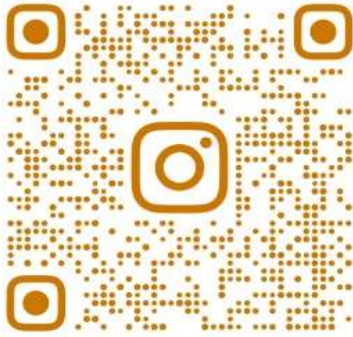
(BE Computer 2019 PAT)

A.Y. 2022-23

Prepared by

Mr. Dhomse G.P.

Join community by clicking below links



@SPPU_ENGINEERING_UPDATE

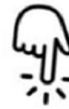


Insta Page

https://www.instagram.com/sppu_engineering_update



@SPPU_TE_BE_COMP



Telegram Channel

https://t.me/SPPU_TE_BE_COMP



WhatsApp Channel

<https://whatsapp.com/channel/0029VaAhRMdAzNbmPG0jyq2x>

Unit-1 Introduction To Machine Learning

- Introduction to Machine Learning, Comparison of Machine learning with traditional programming, ML vs AI vs Data Science. 1 Hr.
- Types of learning: Supervised, Unsupervised, and semi-supervised, reinforcement learning techniques, 1 Hr.
- Models of Machine learning: Geometric model, Probabilistic Models, Logical Models, Grouping and grading models, 2Hr
- Parametric and non-parametric models. 1 Hr.
- Important Elements of Machine Learning- Data formats
- Learnability 1 Hr.
- Statistical learning approaches 1 Hr.

Introduction to Machine Learning

- **Arthur Samuel**, an early American leader in the field of computer gaming and artificial intelligence, coined the term **“Machine Learning”** in **1959** while at IBM.
- He defined machine learning as **“the field of study that gives computers the ability to learn without being explicitly programmed”**. However, there is no universally accepted definition for machine learning.
- Different authors define the term differently.

Introduction to Machine Learning

- Machine learning is **programming computers to optimize a performance criterion using example data or past experience .**
- **We have a model defined up to some parameters,** and learning is the execution of a computer program to **optimize the parameters of the model using the training data or past experience.**
- **The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data.**
- The field of study known as machine learning is concerned with the question of how to construct computer programs that automatically improve with experience.

Introduction to Machine Learning

- **Definition of learning:** A computer program is said to *learn* from experience **E** with respect to **some class of tasks T** and **performance measure P**,
- if its performance at tasks T, as measured by P improves with experience E.
- **Examples Handwriting recognition learning problem**
 - **Task T** : Recognizing and classifying handwritten words within images
 - **Performance P** : Percent of words correctly classified
 - **Training experience E** : A dataset of handwritten words with given classifications

Introduction to Machine Learning

A robot driving learning problem

- **Task T** : Driving on highways using vision sensors
- **Performance P** : Average distance traveled before an error
- **Training experience E** : A sequence of images and steering commands recorded while observing a human driver

• **Definition:** A computer program which learns from experience is called a machine learning program or simply a learning program .

Introduction to Machine Learning

Machine Learning Examples



- *face recognition on your phone or voice understanding,*
- *diagnose diseases by symptoms (Watson),*
- *advise products,*
- *books (Amazon),*
- *movies (Netflix),*
- *music (Spotify),*

Comparison of Machine learning with traditional programming

TRADITIONAL PROGRAMMING



MACHINE LEARNING



- *Traditional programming you **hard code the behavior of the program.***
- *In machine learning, you leave a lot of that to the **machine to learn from data.***

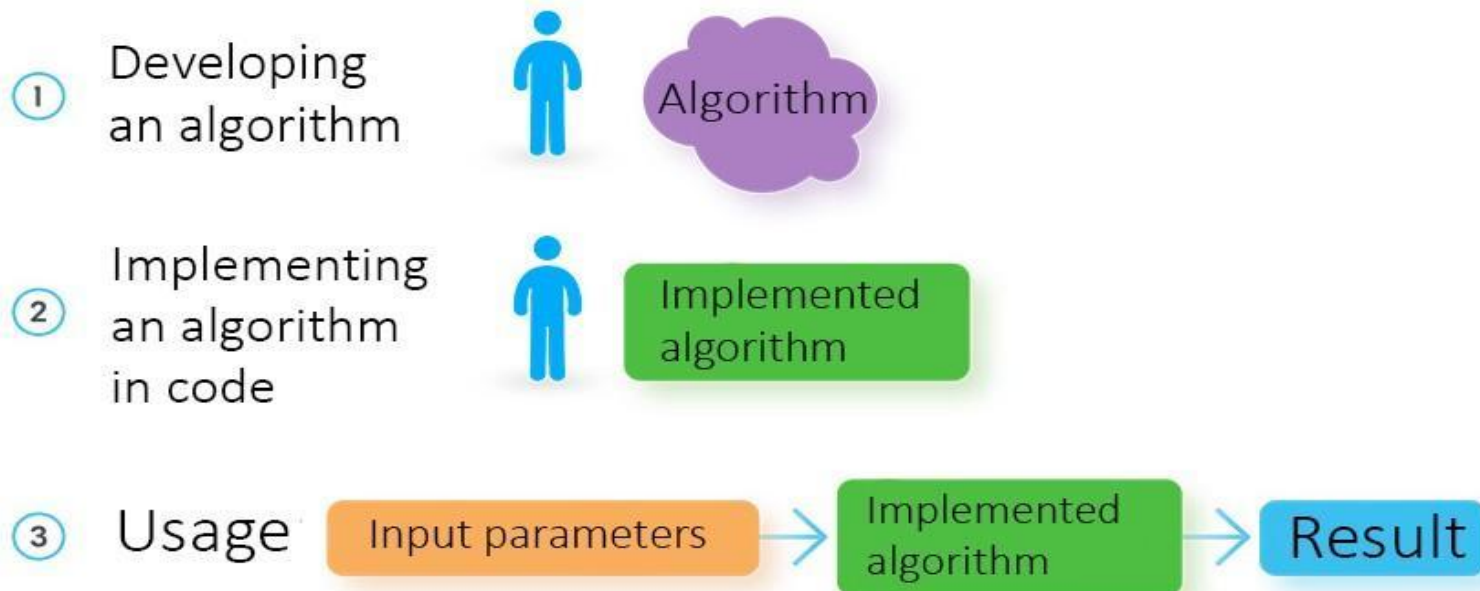
Introduction to Machine Learning

- ML just like AI is not a substitution, but supplementation for traditional programming approaches.
- For instance, ML can be used to build predictive algorithms for an online trading platform, while the platform's UI, data visualization and other elements will be performed in a mainstream programming language such as Ruby, or Java.
- **ML is used in the case when traditional programming strategy falls behind and it is not enough to fully implement a certain task.**

Introduction to Machine Learning

- **Traditional programming approach**

- For any solution, the first task is the creation of the most suitable algorithm and writing the code.
- Thereafter, it is mandatory to set the input parameters and, in fact, if an implemented algorithm is ok it will produce the expected result.

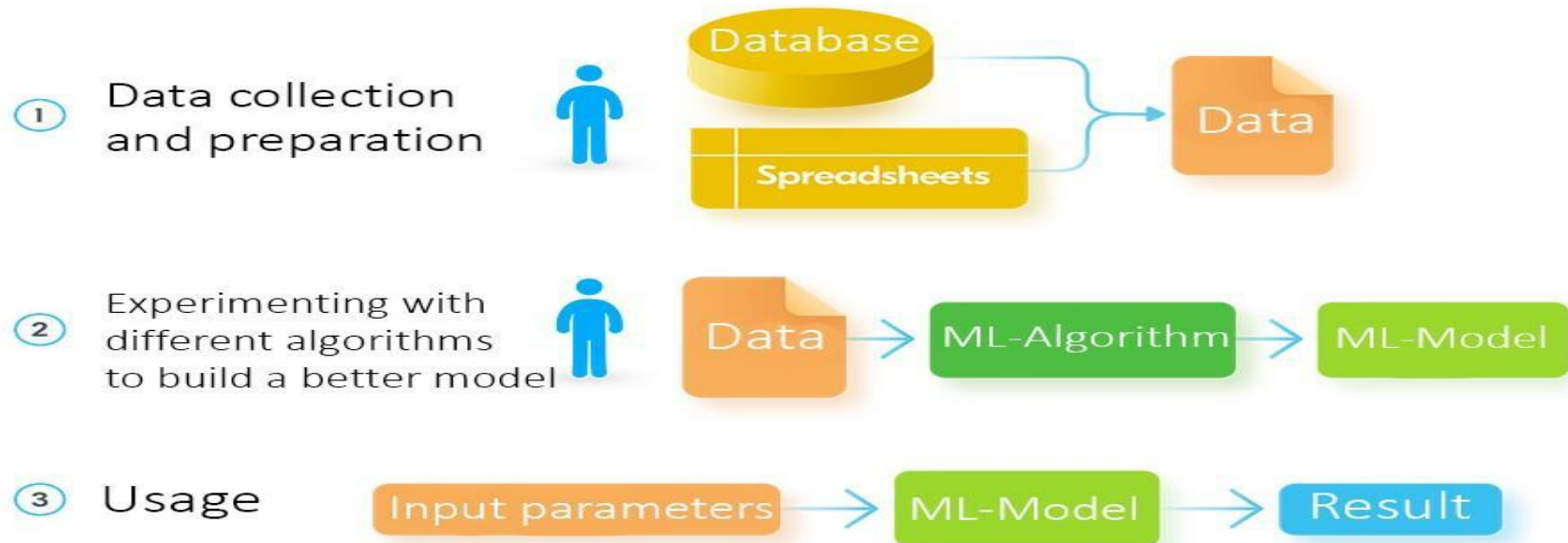


Introduction to Machine Learning

- However, when we need to predict something, we need to use an algorithm with a variety of input parameters.
- In case of prediction of the exchange rate, **it's mandatory to add such details like yesterday's rate; external and internal economic changes in the country that issues the currency and more.**
- Well, it's simple, we need to add a thousand and hundreds of parameters, whereas their limited set allows building a very basic and unscalable model.

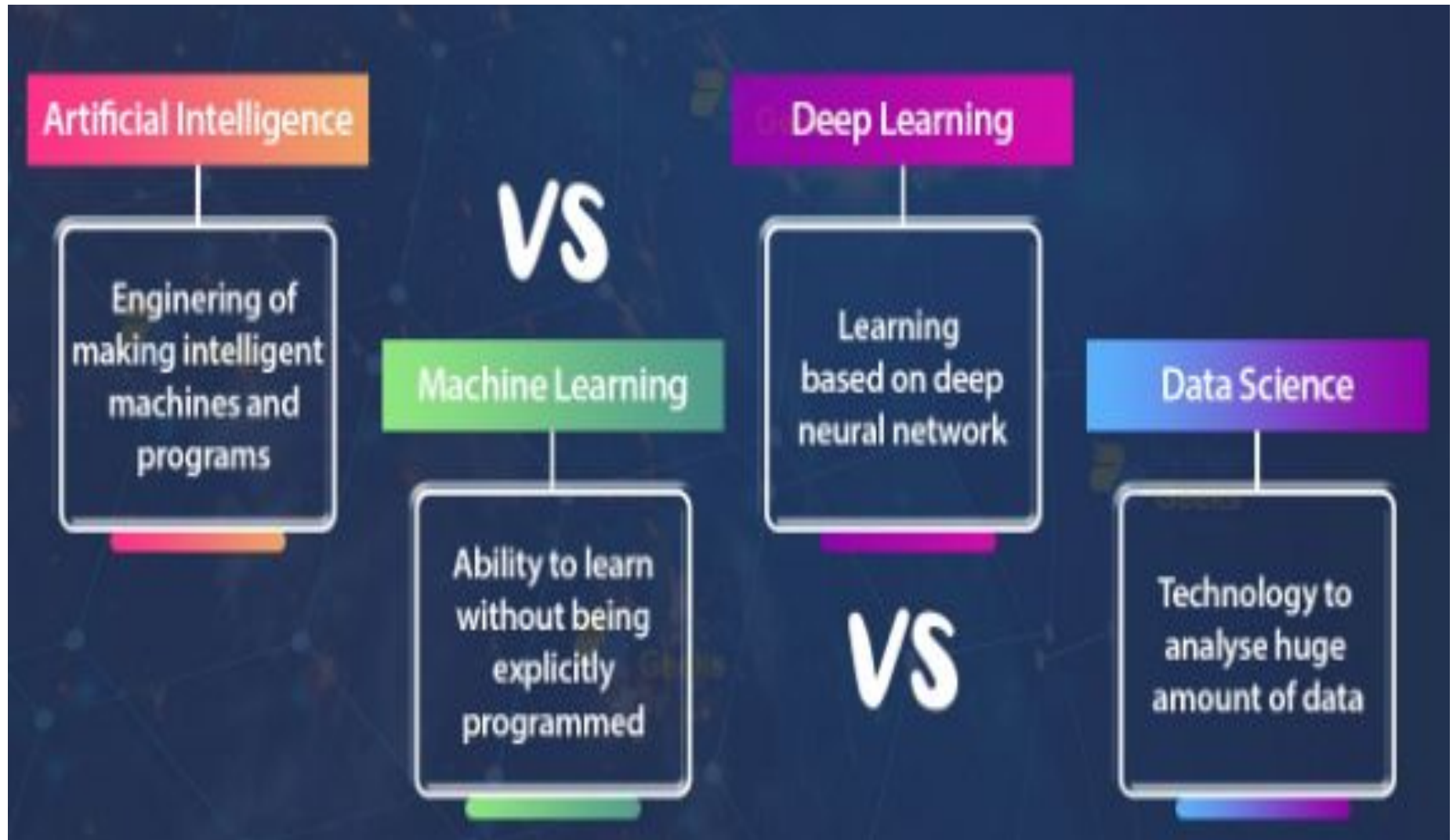
Introduction to Machine Learning

- How a data engineer develops a solution using machine learning <https://morioh.com/p/1063d43ef15e>



- Instead of developing an algorithm on its own, they **need to collect an array of historical data that will be used for semi-automatic model building.**
- managing a satisfactory set of data, the data engineer **loads it into already tailored ML-algorithms.** The result is a model that can **predict a new result, receiving new data as input.**

ML vs AI vs Data Science.

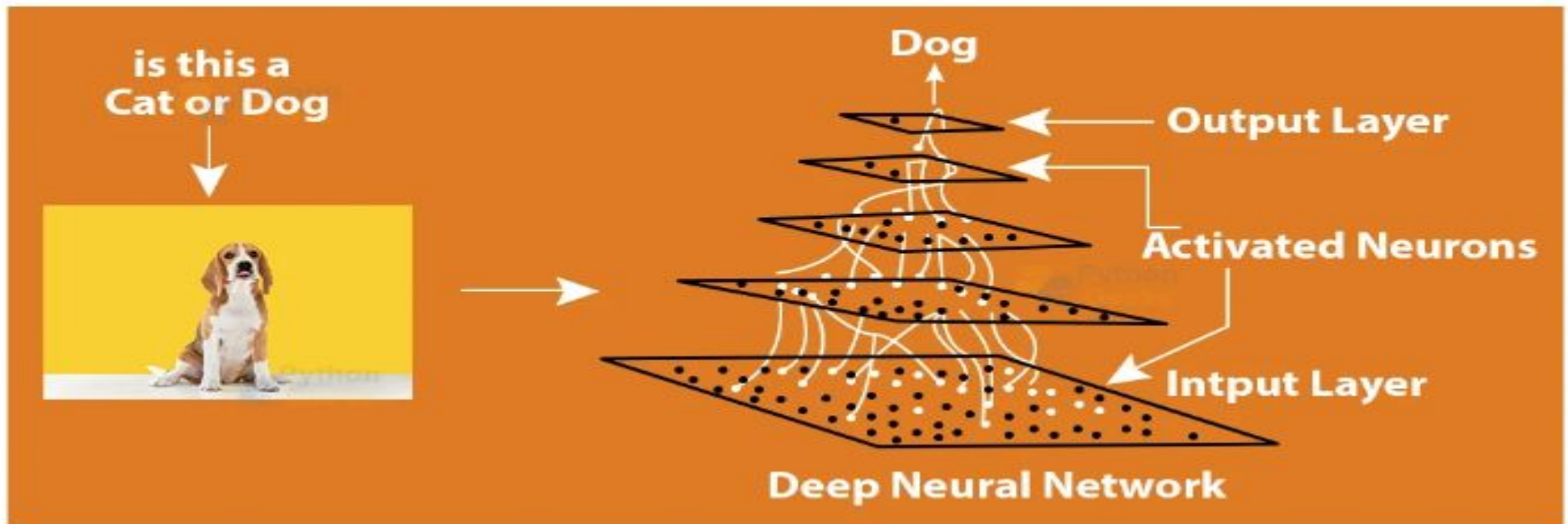


Ref- <https://pythongeeks.org/ai-vs-data-science-vs-deep-learning-vs-ml/>

Introduction to Machine Learning

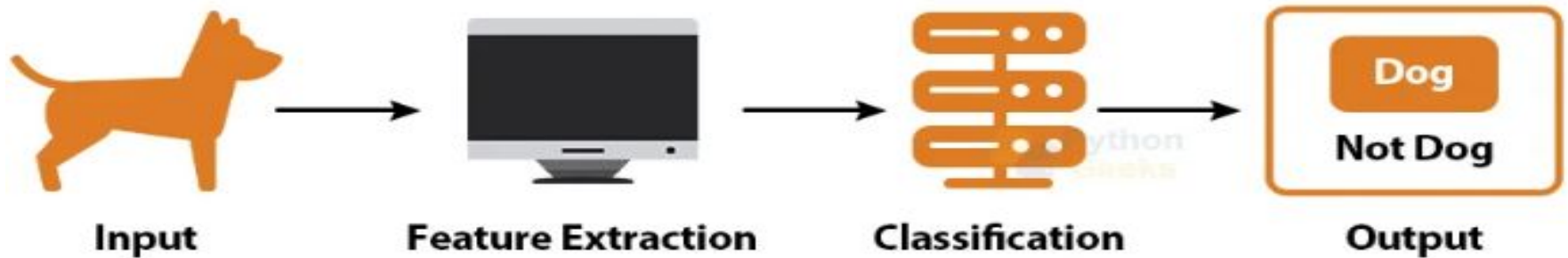
- **Artificial Intelligence-**
- It focuses mainly on building smart machines that are capable of performing tasks that replicate human intelligence without any human interference.
- These systems, built using Artificial Intelligence models, tend to mimic human cognitive functions, that allows decision making, and also helps to improve learning.
- **Example-** forecast financial and business outcomes and can efficiently provide solutions for businesses.

Artificial Intelligence



- We can demonstrate the working of AI in brief with the following steps:
 1. Collect Data
 2. Clean and Prepare Data
 3. Train the Model
 4. Test the Data
 5. Improve

Traditional Machine Learning



- Machine learning algorithms make use of computational methods and try to “learn” from the input data without the requirement of any predetermined equation.
- It comprises an application of AI that allows the systems to learn and improve significantly from the past data experience.

- The **working of the Machine Learning** models is simply put as:
 1. Gather data from source
 2. Clean and filter the data
 3. Choose the effective algorithm according to your problem
 4. Train the test model
 5. Tune in the parameters for best performance
 6. Test the models and try to improve the efficiency
 7. Deploy the final model having precise outputs

Deep Learning

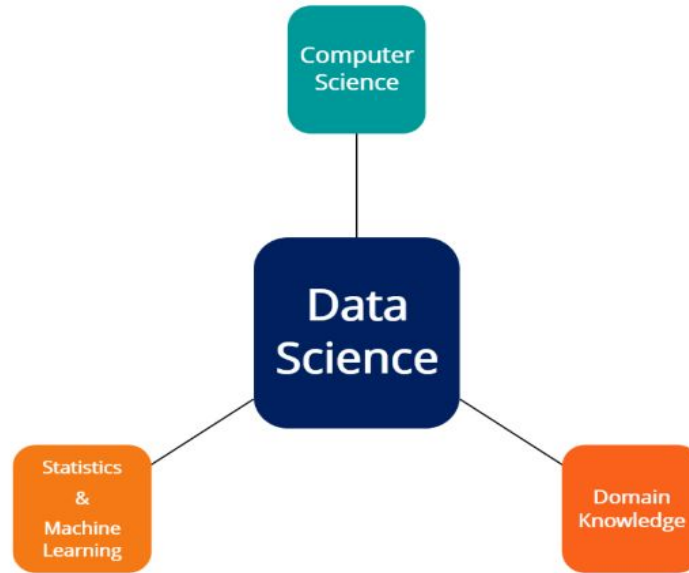


- Deep learning is again a subfield of the artificial intelligence domain.
- It makes use of a multi-layered structure of algorithms, more commonly known as a neural network.
- Deep Learning, like Machine Learning algorithms, also need data for learning and solving problems like classification and prediction.
- We can even consider Deep Learning as a subdomain of machine learning.

DEEP LEARNING

- Unlike Machine Learning, **we do not need trained data** for the model training in Deep Learning, **since we can use this technology when we do not have well classified data.**
- The Deep Learning system looks for appropriate differentiators in the given data points without considering any external classification.
- This is how we can avoid the situation of human interference for the training of the Deep Learning Model.
- **These models analyze new entities for new features at each layer and use it to choose the way in which we can classify the entries.**
- **The system keeps on checking itself** in order to look for new classifications or categories that we can generate from the new entities.

DATA SCIENCE



- The main focus of **data science models** is to **recognize patterns in the given input data sets**.
- **It makes use of numerous statistical techniques to analyze and extract information** like features and classifiers of the data from the given data sets.

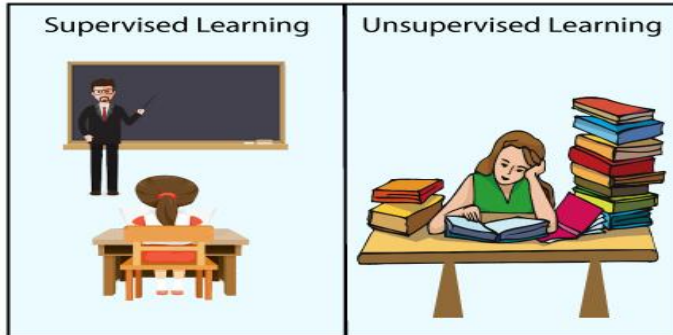
DATA SCIENCE

- With the help of these valuable insights, data scientists can help companies make smarter business decisions based on their data about previous sales and marketing figures.
- We can understand the **working of Data Science** model with the following points in brief:
 1. Gather data from sources
 2. Filter and process the data
 3. Find trends in data and get insight
 4. Build useful data models
 5. Measure the performance

Difference between AI vs Data Science vs Deep Learning vs ML

	Artificial Intelligence	Machine Learning	Deep Learning	Data Science
Technology	Uses decision trees, logic, and statistical data to mimic human intelligence	Uses statistical methods and algorithms to enable machines to learn from experience	Relies on algorithms and artificial neural networks, designed to imitate how humans think and learn	Uses math, programming, and business analysis to produce insights from huge volumes of data
Computational Requirements	Requires more computational power to make machines acquire human-level intelligence	Requires high-performance computers with good-quality GPU for the machine learning algorithms to function	Deep learning is computationally expensive given it is remarkable at modeling diverse phenomena	Requires higher RAM to find and extract patterns in data
Problem Solving Pattern	Chooses algorithms basis complexity of the problem, contributing towards cost and time saving	Divides a given problem into subsets, solves individually, and gives a combined output	Analyzes difficult problems in its hidden layers and performs automatic feature extraction	Collect, analyze, and define data. DS makes use of analytics-based approaches to extract knowledge and actionable insights
Data Dependency	Requires a lot of data to work on and obtain results	The more data a system receives, the more it learns to function better	Powered by massive amounts of data	Heavily dependent on data
Functionality	It accounts for a broader segment and focuses mainly on intelligent behavior than accuracy	Results are largely dependent on accuracy and patterns	Derives specific features from the raw input	Proposes insightful solutions from raw undefined data for effective decision making

Types of -Machine Learning



- **Supervised Machine Learning:**
- **Supervised learning is a machine learning method in which models are trained using labeled data.** In supervised learning, models need to find the mapping function to map the input variable (X) with the output variable (Y).

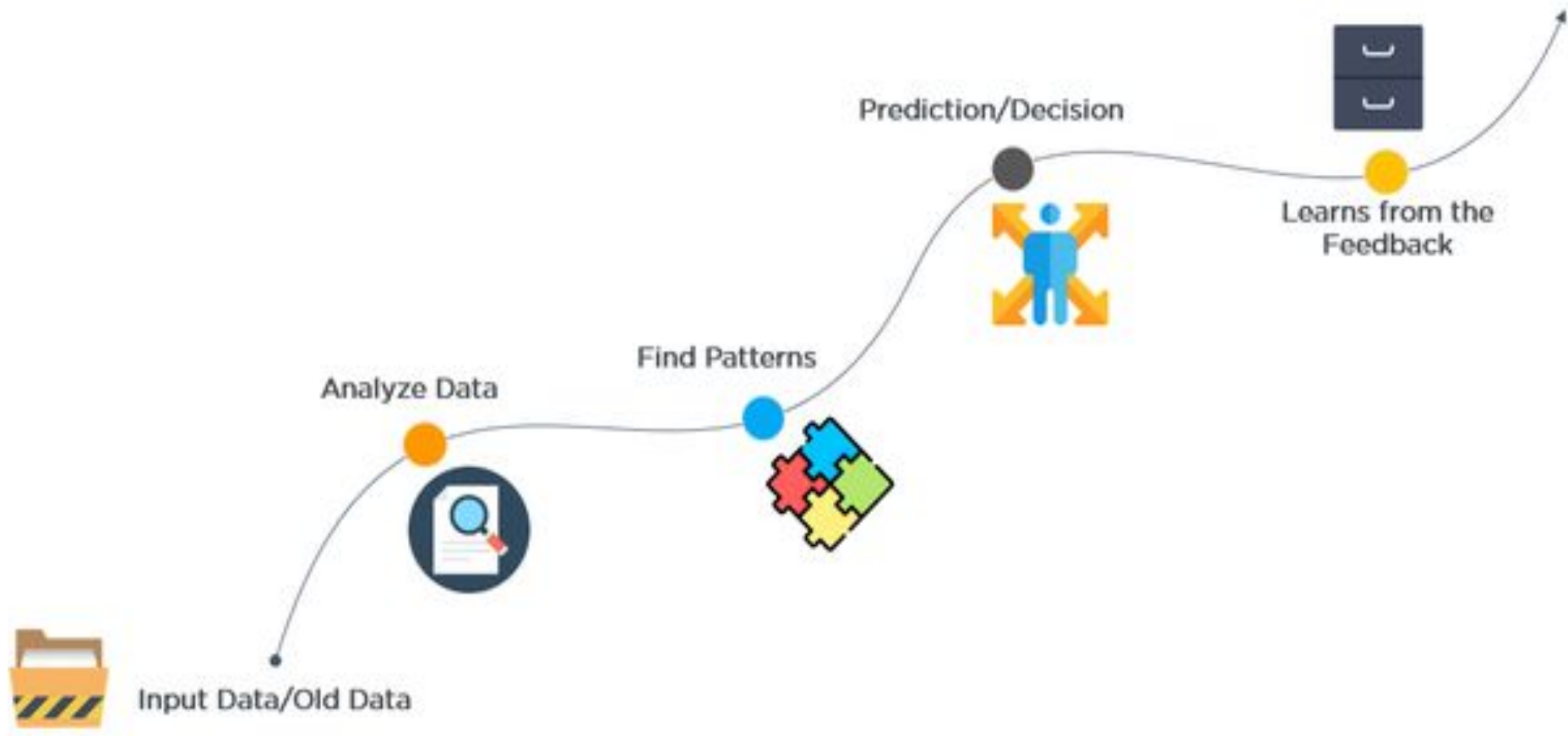
$$Y = f(X)$$

- Supervised learning needs supervision to train the model, which is similar to as a student learns things in the presence of a teacher. Supervised learning can be used for two types of problems: **Classification** and **Regression**.

Types of -Machine Learning

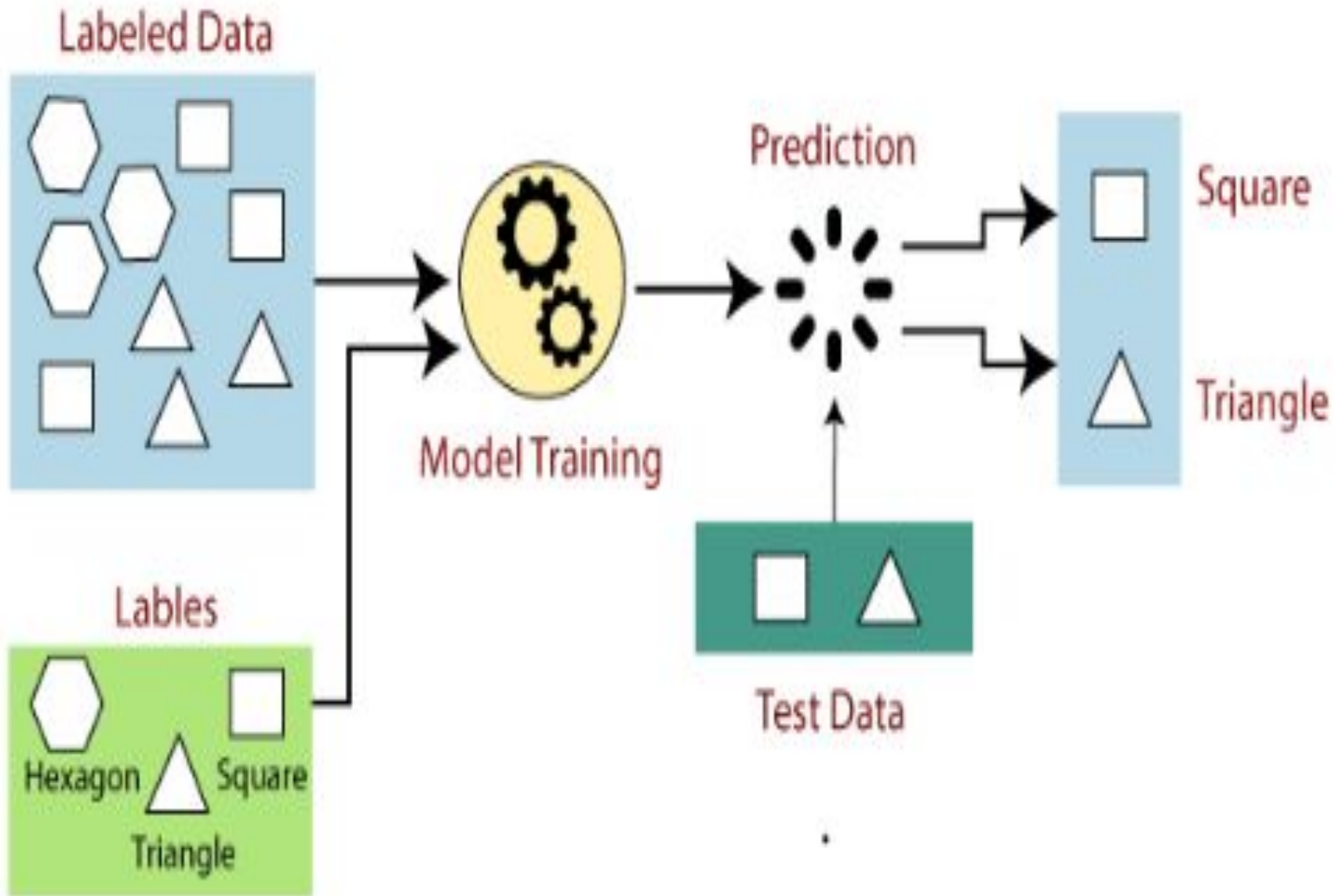
- Let us say we have a **dataset** that contains pictures of different kinds of **fruits** and we want Machine Learning to **segregate the photos based on the kind of fruits**.
- First we provide the dataset to the system i.e we provide the input data.
- The system goes through the entire dataset or analyses **it to find patterns based on size, shapes, colors, etc.**
- Now that it has figured out the patterns, the **systems** takes decisions and starts separating the photos based on the patterns.
- Once the work is done, the **system learns from the feedback it gets**. If it gets any of the fruit type wrong, it will make sure it does not happen in the future.

Supervised Machine Learning



- In the real-world, supervised learning can be used for **Risk Assessment, Image classification, Fraud Detection, spam filtering, etc.**

How Supervised Learning Works?



How Supervised Learning Works?

- Suppose we have a dataset of different types of shapes which includes square, rectangle, triangle, and Polygon. Now the first step is that we need to train the model for each shape.
- If the given shape has four sides, and all the sides are equal, then it will be labelled as a **Square**.
- If the given shape has three sides, then it will be labelled as a **triangle**.
- If the given shape has six equal sides then it will be labelled as **hexagon**.
- Now, after training, we test our model using the test set, and the task of the model is to identify the shape.
- The machine is already trained on all types of shapes, and when it finds a new shape, it classifies the shape on the bases of a number of sides, and predicts the output.

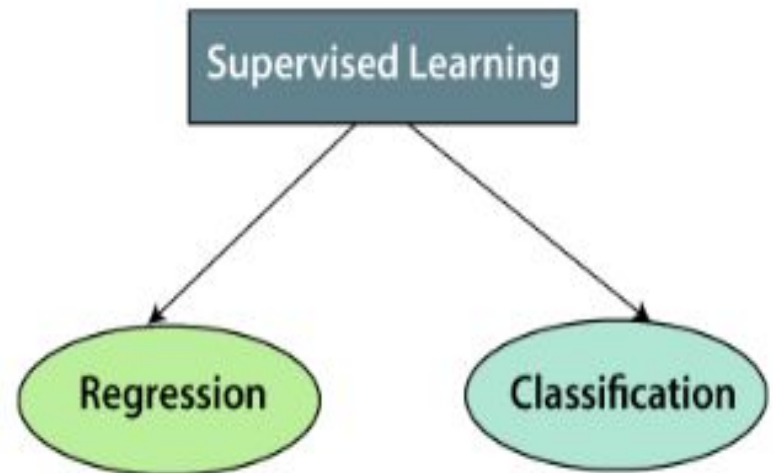
How Supervised Learning Works?

- **Steps Involved in Supervised Learning:**
- First Determine the type of training dataset
- Collect/Gather the labelled training data.
- Split the training dataset into training **dataset**, **test dataset**, and **validation dataset**.
- Determine the input features of the training dataset, which should have enough knowledge so that the model can accurately predict the output.
- Determine the suitable algorithm for the model, such as support vector machine, decision tree, etc.
- Execute the algorithm on the training dataset. Sometimes we need validation sets as the control parameters, which are the subset of training datasets.
- Evaluate the accuracy of the model by providing the test set. If the model predicts the correct output, which means our model is accurate.

Types of supervised Machine learning Algorithms:

1. Regression

- **Regression algorithms are used if there is a relationship between the input variable and the output variable.**
- It is used for the prediction of continuous variables, such as **Weather forecasting, Market Trends, etc.**
- Below are some popular Regression algorithms which come under supervised learning:
- Linear Regression
- Regression Trees
- Non-Linear Regression
- Bayesian Linear Regression
- Polynomial Regression



Supervised Machine Learning

2. Classification

- Classification algorithms are used when the output variable is categorical, which means there are two classes such as **Yes-No, Male-Female, True-false, etc.**
- Spam Filtering,
- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines

Advantages of Supervised learning:

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- In supervised learning, we can have an exact idea about the classes of objects.
- Supervised learning model helps us to solve various real-world problems such as **fraud detection**, **spam filtering**, etc.

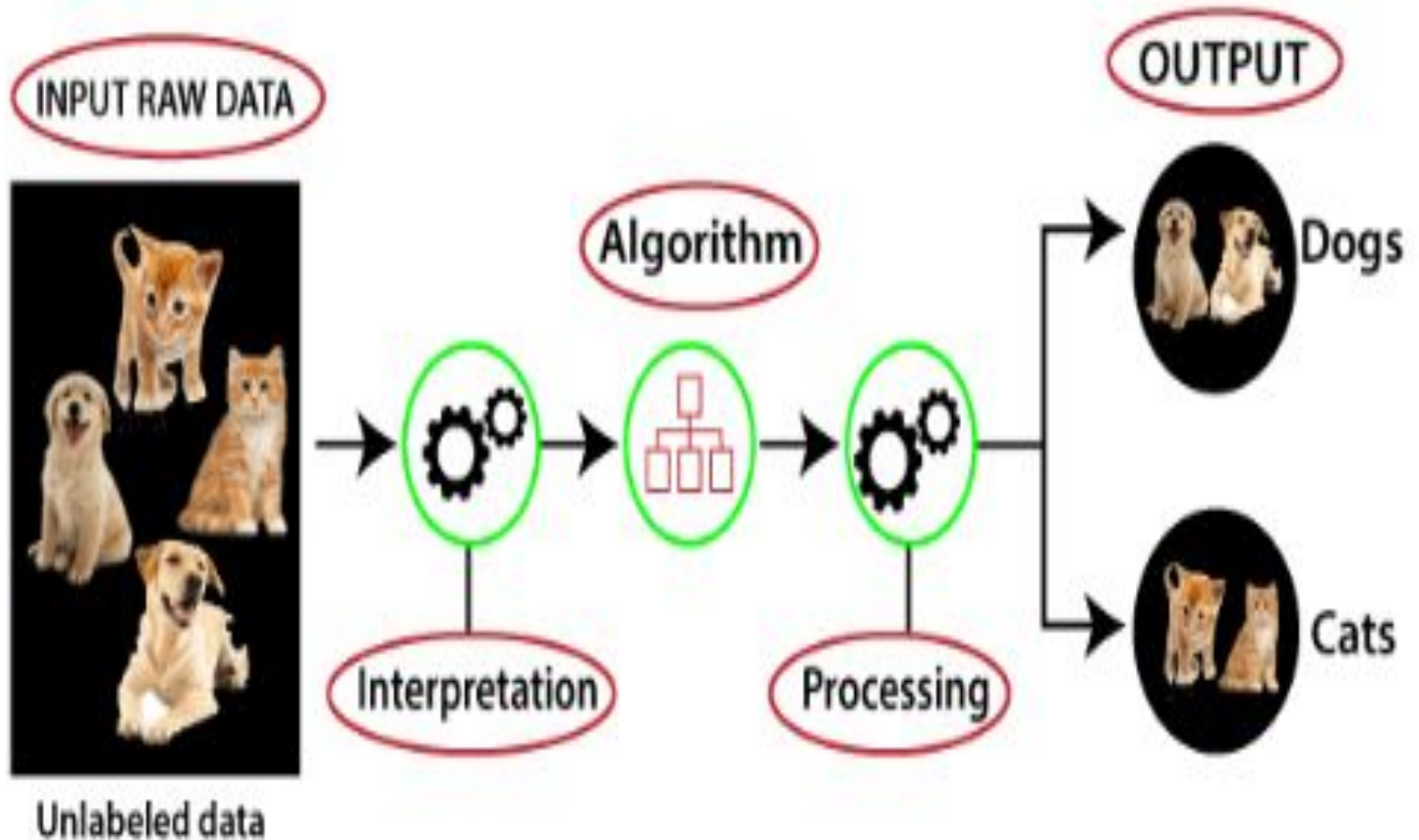
Disadvantages of supervised learning:

- Supervised learning models are not suitable for handling the complex tasks.
- Supervised learning cannot predict the correct output if the test data is different from the training dataset.
- Training required lots of computation times.
- In supervised learning, we need enough knowledge about the classes of object.

Unsupervised Machine Learning

- Unsupervised learning is where you only have **input data (X)** and **no corresponding output variables**.
- As the name suggests, unsupervised learning is a machine learning technique in which models are not supervised using training dataset.
- Instead, models itself find the hidden patterns and insights from the given data. It can be compared to learning which takes place in the human brain
- The goal of unsupervised learning is to **find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format.**

Working of Unsupervised Learning



- Here, we have taken an unlabeled input data, which means it is not categorized and corresponding outputs are also not given.
- Now, this unlabeled input data is fed to the machine learning model in order to train it. Firstly, it will interpret the raw data to find the hidden patterns from the data and then will apply suitable algorithms such as k-means clustering, Decision tree, etc.
- Once it applies the suitable algorithm, the algorithm divides the data objects into groups according to the similarities and difference between the objects.

Types of Unsupervised Learning Algorithm:

- **Clustering:** Clustering is a method of grouping the objects into clusters such that objects with most **similarities remains into a group** and has less or no similarities with the objects of another group.
- **Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.**
- **Association:** An association rule is an unsupervised learning method which is used for **finding the relationships between variables in the large database**. It determines the set of items that occurs together in the dataset.
- Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.

- **Unsupervised Learning algorithms:**
- Below is the list of some popular unsupervised learning algorithms:
- **K-means clustering**
- **KNN (k-nearest neighbors)**
- **Hierarchical clustering**
- **Anomaly detection**
- **Neural Networks**
- **Principle Component Analysis**
- **Independent Component Analysis**
- **Apriori algorithm**
- **Singular value decomposition**

Advantages of Supervised learning:

- Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

Disadvantages of Unsupervised Learning

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.

Difference between Supervised vs Unsupervised Learning

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.

Difference between Supervised vs Unsupervised Learning

Supervised learning can be categorized in **Classification** and **Regression** problems.

Unsupervised Learning can be classified in **Clustering** and **Associations** problems.

Supervised learning can be used for those cases where we know the input as well as corresponding outputs.

Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.

Supervised learning model produces an accurate result.

Unsupervised learning model may give less accurate result as compared to supervised learning.

Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.

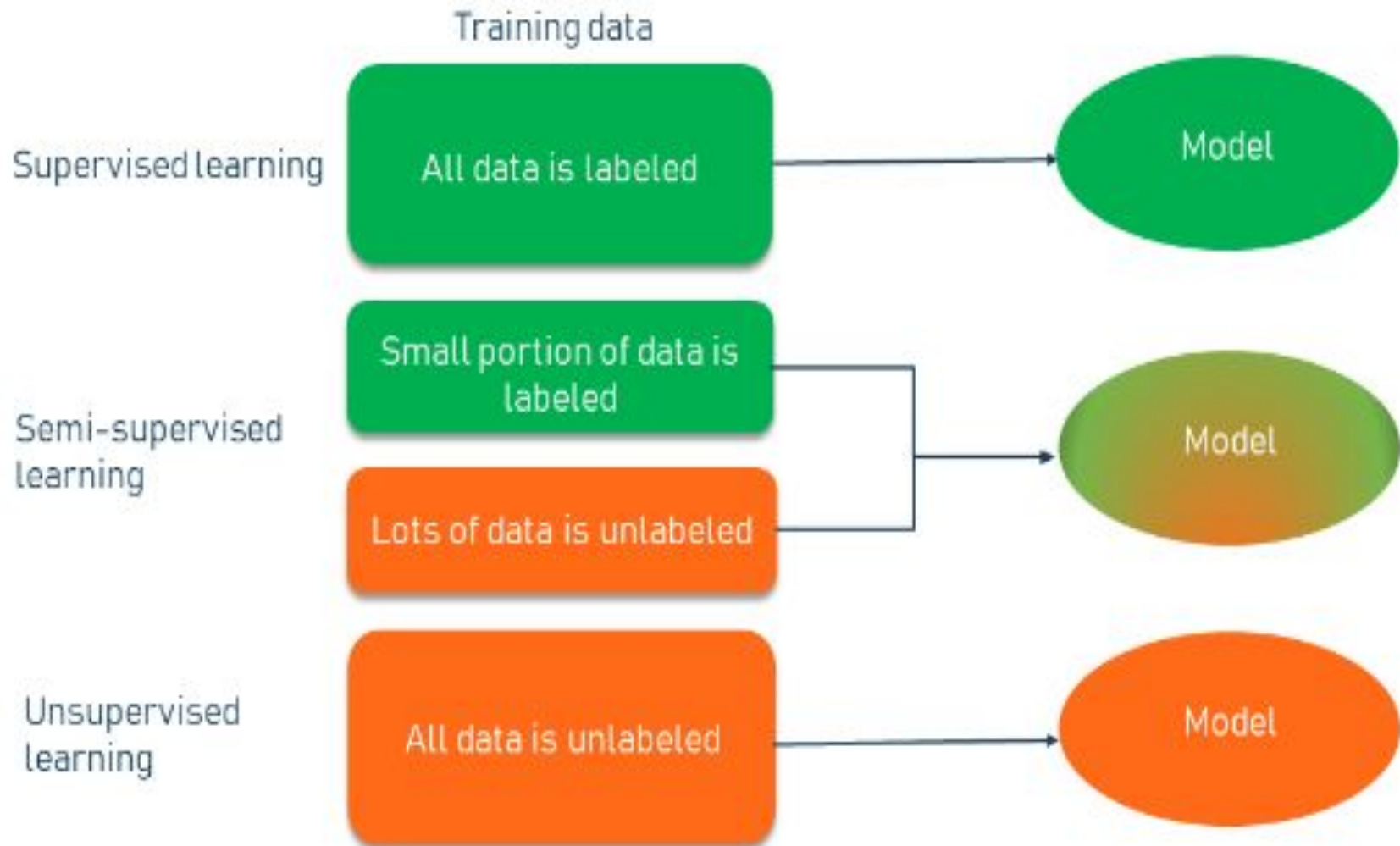
Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.

It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.

It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

Semi-Supervised Learning

SUPERVISED LEARNING vs SEMI-SUPERVISED LEARNING vs UNSUPERVISED LEARNING



Semi-supervised learning

- **Semi-supervised learning** bridges supervised learning and unsupervised learning techniques to solve their key challenges.
- With it, **you train an initial model on a few labeled samples and then iteratively apply it to the greater number of unlabeled data.**
- Unlike unsupervised learning, **SSL works for a variety of problems from classification and regression to clustering and association.**
- Unlike supervised learning, **the method uses small amounts of labeled data and also large amounts of unlabeled data**, which reduces expenses on manual annotation and cuts data preparation time.

How Semi-supervised learning work

SEMI-SUPERVISED SELF-TRAINING METHOD



- You pick a small amount of labeled data, e.g., images showing cats and dogs with their respective tags, and you use this dataset to train a base model with the help of ordinary supervised methods.
- Then you apply the process known as **pseudo-labeling** — when you take the **partially trained model and use it to make predictions for the rest of the database which is yet unlabeled**. The labels generated thereafter are called *pseudo* as they are produced based on the **originally labeled data that has limitations** (say, there may be an uneven representation of classes in the set resulting in bias — more dogs than cats).
- From this point, you take the most confident predictions made with your model (for example, **you want the confidence of over 80 percent that a certain image shows a cat, not a dog**). If any of the pseudo-labels exceed this confidence level, you add them into the **labeled dataset** and create a new, combined input to train an improved model.
- **The process can go through several iterations (10 is often a standard amount)** with more and more pseudo-labels being added every time. Provided the data is suitable for the process, the performance of the

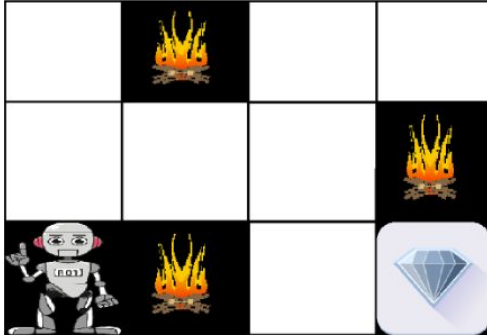
Semi-supervised learning examples

- **Speech Recognition-** Facebook (now Meta) has successfully applied semi-supervised learning (namely the self-training method) to its speech recognition models and improved them.
- **Web content classification-** Many search engines, including Google, apply SSL to their ranking component to better understand human language and the relevance of candidate search results to queries.
- **Text document classification-** building of a text document classifier.

Reinforcement Learning Techniques

- Reinforcement learning is an area of Machine Learning. **It is about taking suitable action to maximize reward in a particular situation.**
- in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task. In the absence of a training dataset, it is bound to learn from its experience.
- **Example:** The problem is as follows: We have an agent and a reward, with many hurdles in between. The agent is supposed to find the best possible path to reach the reward. The following problem explains the problem more easily.

Reinforcement Learning Techniques



- **The above image shows the robot, diamond, and fire. The goal of the robot is to get the reward that is the diamond and avoid the hurdles that are fired.**
- The robot learns by trying all the possible paths and then choosing the path which gives him the reward with the least hurdles.
- **Each right step will give the robot a reward and each wrong step will subtract the reward of the robot.** The total reward will be calculated when it reaches the final reward that is the diamond.

Reinforcement Learning Techniques

- **Main points in Reinforcement learning –**
 - Input:** The input should be an initial state from which the model will start
 - Output:** There are many possible outputs as there are a variety of solutions to a particular problem
 - Training:** The training is based upon the input, The model will return a state and the user will decide to reward or punish the model based on its output.
 - The model keeps continues to learn.
- **The best solution is decided based on the maximum reward.**

Difference

Reinforcement learning

Reinforcement learning is all about making decisions sequentially. In simple words, we can say that the output depends on the state of the current input and the next input depends on the output of the previous input

In Reinforcement learning decision is dependent, So we give labels to sequences of dependent decisions

Example: Chess game

Supervised learning

In Supervised learning, the decision is made on the initial input or the input given at the start

In supervised learning the decisions are independent of each other so labels are given to each decision.

Example: Object recognition

Negative Reinforcement

Before	Behaviour	After
Bad smell	Have a shower	Bad smell gone

Positive Reinforcement

Before	Behaviour	After
No hot water	Turn on hot water tap (faucet)	Hot water added

Types of Reinforcement:

- There are two types of Reinforcement:

- **Positive**

Positive Reinforcement is defined as **when an event, occurs due to a particular behavior**, increases the strength and the frequency of the behavior. In other words, it has a positive effect on behavior.

- **Advantages of reinforcement learning are:**

- Maximizes Performance
- Sustain Change for a long period of time
- Too much Reinforcement can lead to an overload of states which can diminish the results

Types of Reinforcement:

- **Negative**

Negative Reinforcement is defined as **strengthening of behavior because a negative condition is stopped or avoided.**

- **Advantages of reinforcement learning:**

- Increases Behavior
- Provide defiance to a minimum standard of performance
- It Only provides enough to meet up the minimum behavior

- **State-Action-Reward-State-Action (SARSA):** SARSA is an On-policy algorithm based on the Markov decision process. It uses the action performed by the current policy to learn the Q-value. The SARSA algorithm stands **for State Action Reward State Action, which symbolizes the tuple (s, a, r, s', a') .**



Practical applications Reinforcement Learning

- RL can be used in **robotics for industrial automation**.
- RL can be used in **machine learning and data processing**
- RL can be used to create training systems that provide custom instruction and materials according to the requirement of students.
- RL can be used in large environments in the following situations:
A model of the environment is known, but an analytic solution is not available;
- Only a simulation model of the environment is given (the subject of simulation-based optimization)
- The only way to collect information about the environment is to interact with it.

Models of Machine learning: Geometric model

- Some broad categories of models:

1. Geometric models

- E.g. K-nearest neighbors, linear regression, support vector machine, logistic regression, ...

2. Probabilistic models

- Naïve Bayes, Gaussian process regression, conditional random field, ...

3. Logical models

- Decision tree, random forest, ... Compositional models
Neural networks, logistic regression, ..

4. Ensemble models- Boosting, bagging, random forest.

5. Grading vs grouping models

Models of Machine learning: Geometric model

- Machine learning is concerned with using the right features to build the right models that achieve the right tasks.
- The basic idea of Learning models has **divided into three categories.**
- For a given problem, the collection of all possible outcomes represents the sample space or instance space.
- **Using a Logical expression. (Logical models)**
- **Using the Geometry of the instance space. (Geometric models)** instance space is typically just the domain of X (your sample population) and an instance of that space is a particular x with various features. For example, X may be the set of all people and an instance of x will be an individual person who is 30 female and medium heights etc.
- **Using Probability to classify the instance space. (Probabilistic models)**
- **Grouping and Grading**

Logical models

- **Logical models** use a logical expression to divide the instance space into segments and hence construct grouping models.
- A **logical expression** is an expression that returns a Boolean value, i.e., a True or False outcome.
- Once the data is grouped using a logical expression, the data is divided into homogeneous groupings for the problem we are trying to solve.
- For example, for a classification problem, all the instances in the group belong to one class.
- There are mainly two kinds of logical models: **Tree models** and **Rule models**.

Logical models

- **Rule models** consist of a collection of implications or **IF-THEN rules**.
- **Tree-based models**, the **'if-part'** defines a segment and the **'then-part'** defines the behavior of the model for this segment.
- Rule models follow the same reasoning.
- Tree models can be seen as a particular type of rule model where the if-parts of the rules are organized in a tree structure.
- Both Tree models and Rule models use the same approach to supervised learning.

Logical models

- **Logical models and Concept learning**
- To understand logical models further, we need to understand the idea of Concept Learning.
- **Concept Learning involves learning logical expressions or concepts from examples.**
- Concept learning forms the basis of both tree-based and rule-based models.
- More formally, Concept Learning involves acquiring the definition of a general category from a given set of positive and negative training examples of the category.
- **A Formal Definition for Concept Learning is “The inferring of a Boolean-valued function from training examples of its input and output.”**
- In concept learning, we only learn a description for the positive class and label everything that doesn't satisfy that description as negative.

Logical models

A Concept Learning Task – Enjoy Sport Training Examples

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	YES
2	Sunny	Warm	High	Strong	Warm	Same	YES
3	Rainy	Cold	High	Strong	Warm	Change	NO
4	Sunny	Warm	High	Strong	Warm	Change	YES

ATTRIBUTES

CONCEPT

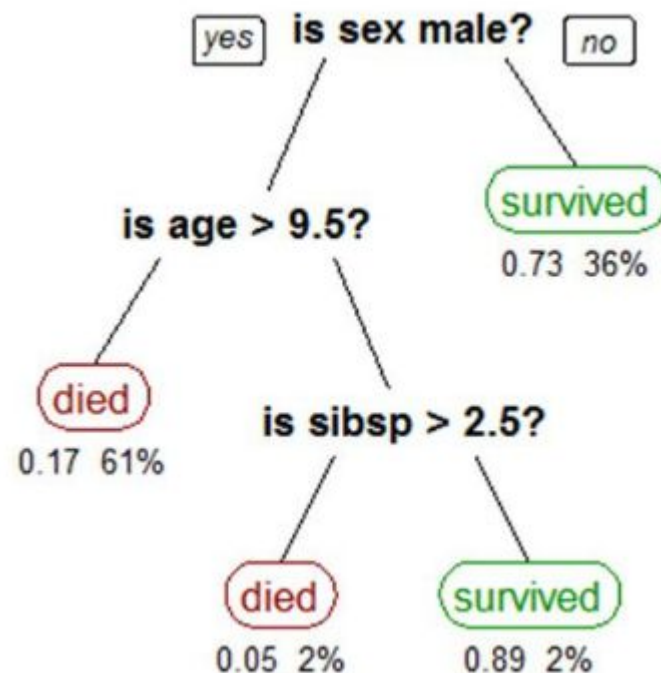
- **The problem can be represented by a series of hypotheses.** Each hypothesis is described by a conjunction of constraints on the attributes. The training data represents a set of positive and negative examples of the target function. **In the example above, each hypothesis is a vector of six constraints, specifying the values of the six attributes – Sky, AirTemp, Humidity, Wind, Water, and Forecast.**
- The training phase involves learning the set of days (as a conjunction of attributes) for which Enjoy Sport = yes.

Models of Machine learning: Geometric model

- Given instances X which represent a set of all possible days, each described by the attributes:
 1. Sky – (values: Sunny, Cloudy, Rainy),
 2. AirTemp – (values: Warm, Cold),
 3. Humidity – (values: Normal, High),
 4. Wind – (values: Strong, Weak),
 5. Water – (values: Warm, Cold),
 6. Forecast – (values: Same, Change).
- **Try to identify a function that can predict the target variable Enjoy Sport as yes/no, i.e., 1 or 0.**

Models of Machine learning: Geometric model

- The tree shows survival numbers of passengers on the Titanic
- ("sibsp" is the number of spouses or siblings aboard).
- The values under the leaves show the probability of survival and the percentage of observations in the leaf.
- The model can be summarised as: Your chances of survival were good if you were (i) a female or (ii) a male younger than 9.5 years with less than 2.5 siblings.



Geometric model

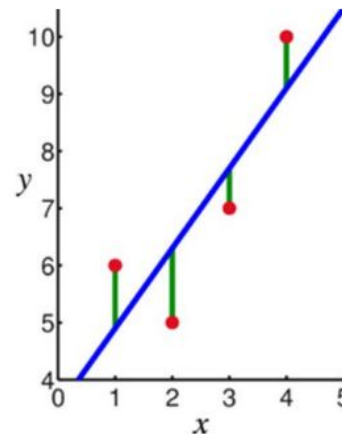
- we have seen that with logical models, **such as decision trees, a logical expression is used to partition the instance space.**
- Two instances are similar when they end up in the same logical segment.
- In this section, **we consider models that define similarity by considering the geometry of the instance space.**
- In Geometric models, features could be described as points in **two dimensions (x- and y-axis)** or a **three-dimensional space (x, y, and z).**
- Eg. , temperature as a function of time can be modelled in two axes)

Geometric model

- There are two ways we could impose similarity.
- **We could use geometric concepts like lines or planes to segment (classify) the instance space. These are called Linear models.**
- Alternatively, we can use the geometric notion of distance to represent similarity.
- **In this case, if two points are close together, they have similar values for features and thus can be classed as similar. We call such models as Distance-based models.**

Geometric model

- **1. Linear models** are relatively simple.
- In this case, the function is represented as a linear combination of its inputs.
- Thus, if x_1 and x_2 are two scalars or vectors of the same dimension and a and b are arbitrary scalars, then $ax_1 + bx_2$ represents a linear combination of x_1 and x_2 .
- In the simplest case where $f(x)$ represents a straight line, we have an equation of the form
- $f(x) = mx + c$ where c represents the intercept and m represents the slope.

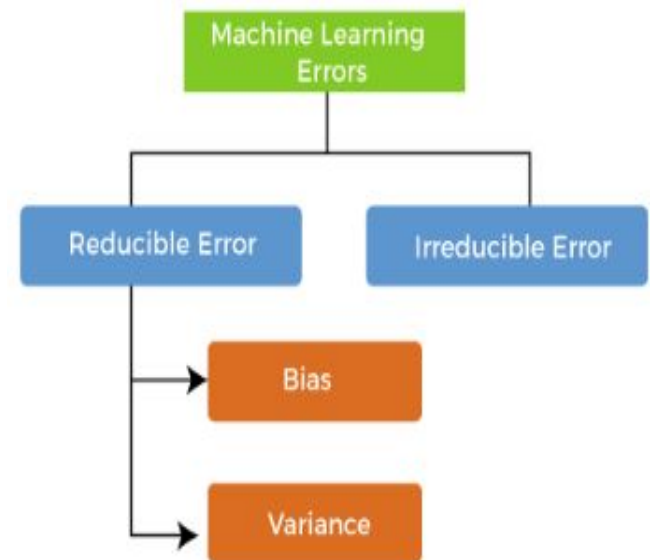


Geometric model

- Linear models are **parametric**, which means that they have a **fixed form with a small number of numeric parameters** that need to be learned from data.
- For example, in $f(x) = mx + c$, m and c are the **parameters that we are trying to learn from the data**.
- Linear models are **stable**, i.e., small variations in the training data have only a limited impact on the learned model.
- In contrast, **tree models tend to vary more with the training data**, as the choice of a different split at the root of the tree typically means that the rest of the tree is different as well.

Geometric model

- Linear models have **low variance and high bias**.
- This implies that **Linear models are less likely to overfit the training data** than some other models.
- However, they are more likely to **underfit**. For example, if we want to learn the boundaries between countries based on labelled data, then linear models are not likely to give a good approximation
- **Errors in Machine Learning?**
- **Reducible errors:** These errors
- can be reduced to improve the
- model accuracy.
- **Irreducible errors:** These errors will
- always be present in the model



Geometric model

- **What is Bias?**

- In general, a machine learning model analyses the data, find patterns in it and make predictions.
- While training, the model learns these patterns in the dataset and applies them to test data for prediction.
- *While making predictions, a difference occurs between prediction values made by the model and actual values/expected values, and this difference is known as bias errors or Errors due to bias.*
- **Low Bias:** A low bias model **will make fewer assumptions** about the form of the target function.
- **High Bias:** A model with a **high bias makes more assumptions**, and the model becomes unable to capture the important features of our dataset. **A high bias model also cannot perform well on new data.**

Geometric model

- **Some examples of machine learning algorithms with low bias** are **Decision Trees, k-Nearest Neighbours and Support Vector Machines**. At the same time, an algorithm with high bias is **Linear Regression, Linear Discriminant Analysis and Logistic Regression**.
- **Ways to reduce High Bias:**
 - High bias mainly occurs due to a much simple model. Below are some ways to reduce the high bias:
 - Increase the input features as the model is underfitted.
 - Use more complex models, such as including some polynomial features.

Geometric model

- **What is a Variance Error?**
- The variance would specify the amount of variation in the prediction if the different training data was used.
- In simple words, *variance tells that how much a random variable is different from its expected value.*
- Ideally, a model should not vary too much from one training dataset to another, which means the algorithm should be good in understanding the hidden mapping between inputs and output variables.
- Variance errors are either of **low variance or high variance.**
- **Low variance** means there is a **small variation in the prediction** of the target function with changes in the training data set. At the same time, **High variance** shows a **large variation in the prediction** of the target function with changes in the training dataset.

Models of Machine learning: Geometric model

- A model that shows high variance learns a lot and perform well with the training dataset, and does not generalize well with the unseen dataset.
- As a result, such a model gives good results with the training dataset **but shows high error rates on the test dataset.**
- Since, with high variance, the model learns too much from the dataset, it leads to overfitting of the model.
- A model with high variance has the below problems:
- A high variance model leads to overfitting.
- Increase model complexities.

Models of Machine learning: Geometric model



- Some examples of machine learning algorithms with low variance are, **Linear Regression, Logistic Regression, and Linear discriminant analysis.**
- At the same time, algorithms with high variance are **decision tree, Support Vector Machine, and K-nearest neighbours.**

Models of Machine learning: Geometric model

Ways to Reduce High Variance:

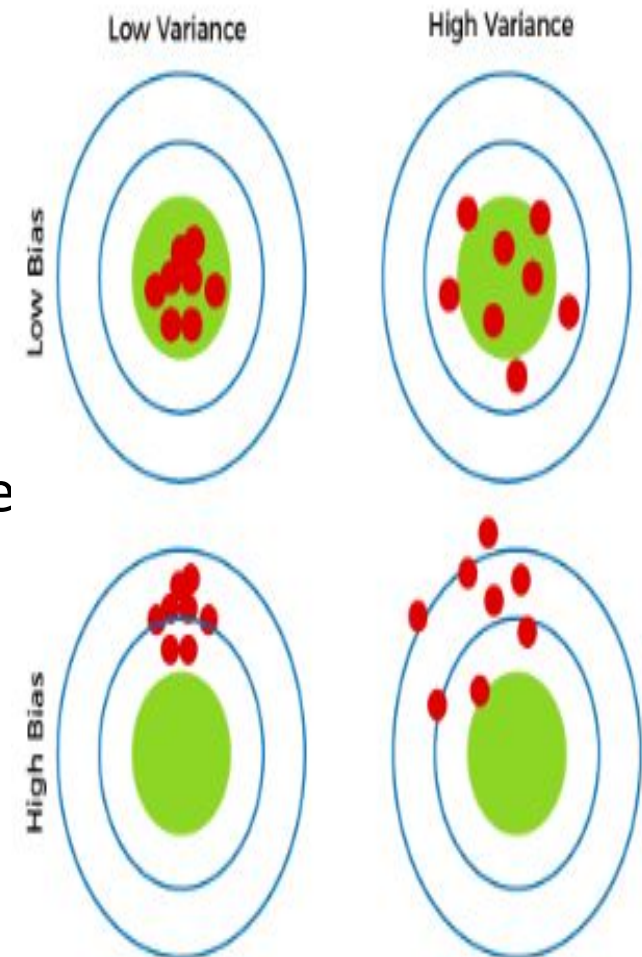
- Reduce the input features or number of parameters as a model is overfitted.
- Do not use a much complex model.
- Increase the training data.
- Increase the Regularization term.

Different Combinations of Bias-Variance

Low-Bias, Low-Variance:

The combination of low bias and low variance shows an ideal machine learning model.

However, it is not possible practically.

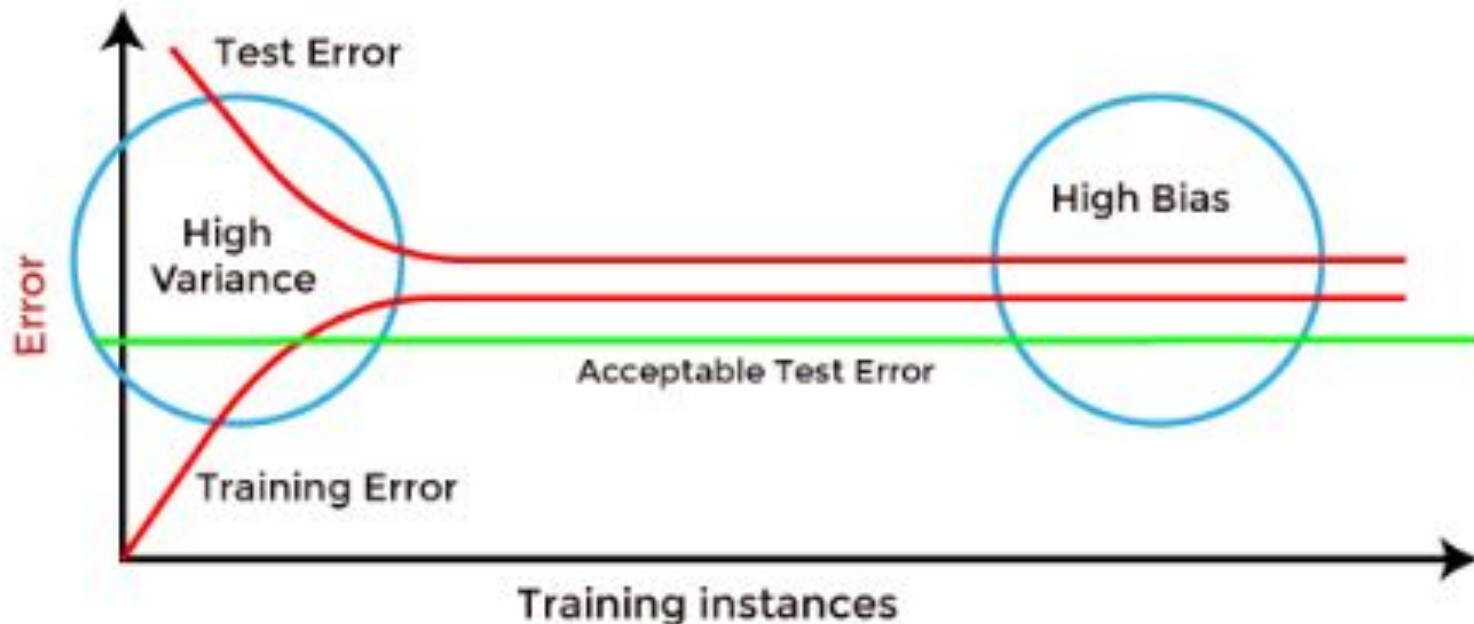


Models of Machine learning: Geometric model

- **Low-Bias, High-Variance:** With low bias and high variance, model predictions are inconsistent and accurate on average. **This case occurs when the model learns with a large number of parameters** and hence leads to an **overfitting**
- **High-Bias, Low-Variance:** With High bias and low variance, predictions are consistent but inaccurate on average. **This case occurs when a model does not learn well with the training dataset or uses few numbers of the parameter.** It leads to **underfitting** problems in the model.
- **High-Bias, High-Variance:**
With high bias and high variance, predictions are inconsistent and also inaccurate on average.

How to identify High variance or High Bias?

High variance can be identified if the model has:



- Low training error and high test error.

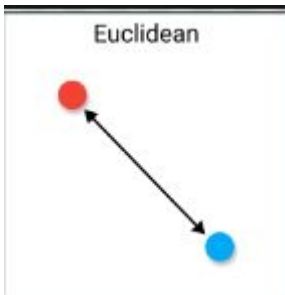
High Bias can be identified if the model has:

- High training error and the test error is almost similar to training error.

Geometric model

- **2. Distance-based models** Distance-based models are the second class of Geometric models.
- Like Linear models, **distance based models are based on the geometry of data.**
- As the name implies, distance-based models
- In the context of Machine learning, the **concept of distance is not based on merely the physical distance between two points.**
- Instead, we could think of the distance between two points considering the mode of transport between two points.
- Travelling between two cities by plane covers less distance physically than by train because a plane is unrestricted.

Models of Machine learning: Geometric model



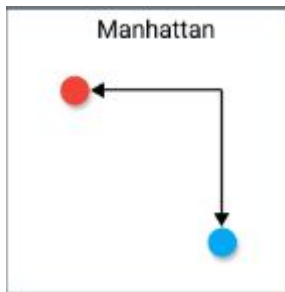
length of a segment connecting two points.

Useful for Less dimensionality

increases of your data □ might be skewed

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Euclidean distance



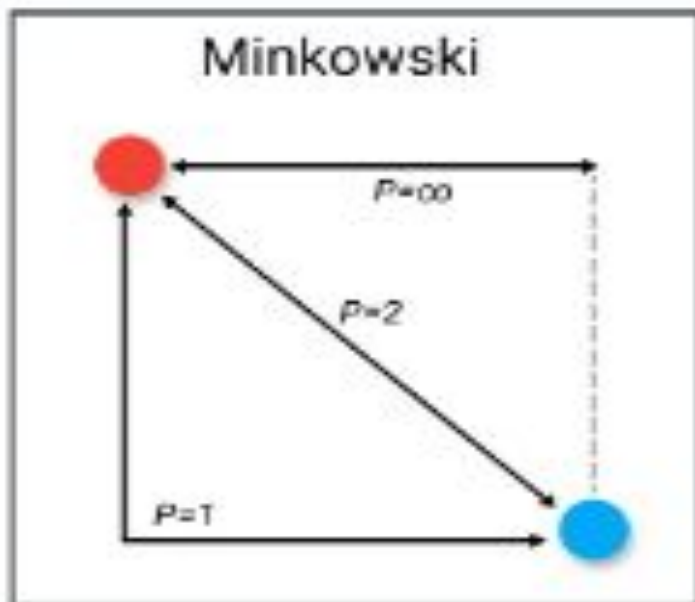
Distance measure on chess board

distance between two vectors

if they could only move right angles

$$D(x, y) = \sum_{i=1}^k |x_i - y_i|$$

Manhattan distance



$$D(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Minkowski distance

Minkowski distance

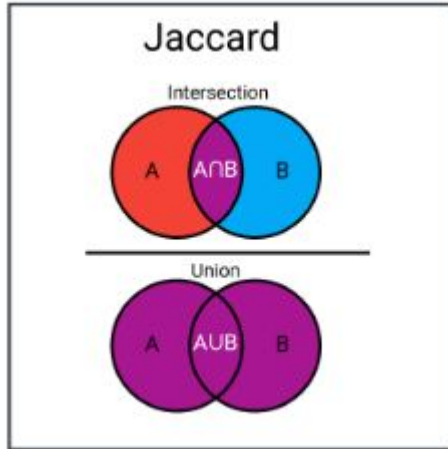
- In Minkowski distance a metric used in Normed vector space (n-dimensional real space), which means that it can be used in a space where distances can be represented as a vector that has a length.
- This measure has three requirements:
- **Zero Vector** — The zero vector has a length of zero whereas every other vector has a positive length. For example, if we travel from one place to another, then that distance is always positive. However, if we travel from one place to itself, then that distance is zero.
- **Scalar Factor** — When you multiple the vector with a positive number its length is changed whilst keeping its direction. For example, if we go a certain distance in one direction and add the same distance, the direction does not change.
- **Triangle Inequality** — The shortest distance between two points is a straight line.

Models of Machine learning: Geometric model

- Most interestingly about this distance measure is the use of parameter p .
- We can use this parameter to manipulate the distance metrics to closely resemble others.
- Common values of p are:
 - $p=1$ — Manhattan distance
 - $p=2$ — Euclidean distance
 - $p=\infty$ — Chebyshev distance

Chebyshev distance- it is simply the maximum distance along one axis. Due to its nature, it is often referred to as **Chessboard distance** since the minimum number of moves needed by a king to go from one square to another is equal to Chebyshev distance.

Models of Machine learning: Geometric model



$$D(x, y) = 1 - \frac{|x \cap y|}{|y \cup x|}$$

- **The Jaccard index (or Intersection over Union) is a metric used to calculate the similarity and diversity of sample sets.** It is the size of the intersection divided by the size of the union of the sample sets.
- In practice, it is the total number of similar entities between sets divided by the total number of entities. **For example, if two sets have 1 entity in common and there are 5 different entities in total, then the Jaccard index would be $1/5 = 0.2$.**

Geometric model

- **Examples of distance-based models** include the **nearest-neighbour** models, which use the training data as exemplars – for example, in classification. The **K-means clustering** algorithm also uses exemplars to create clusters of similar data points.
- **Medoids** are similar in concept to means or centroids. Medoids are most commonly used on data when a mean or centroid cannot be defined.
- They are used in contexts where the centroid is not representative of the dataset, such as in image data.

Probabilistic Models

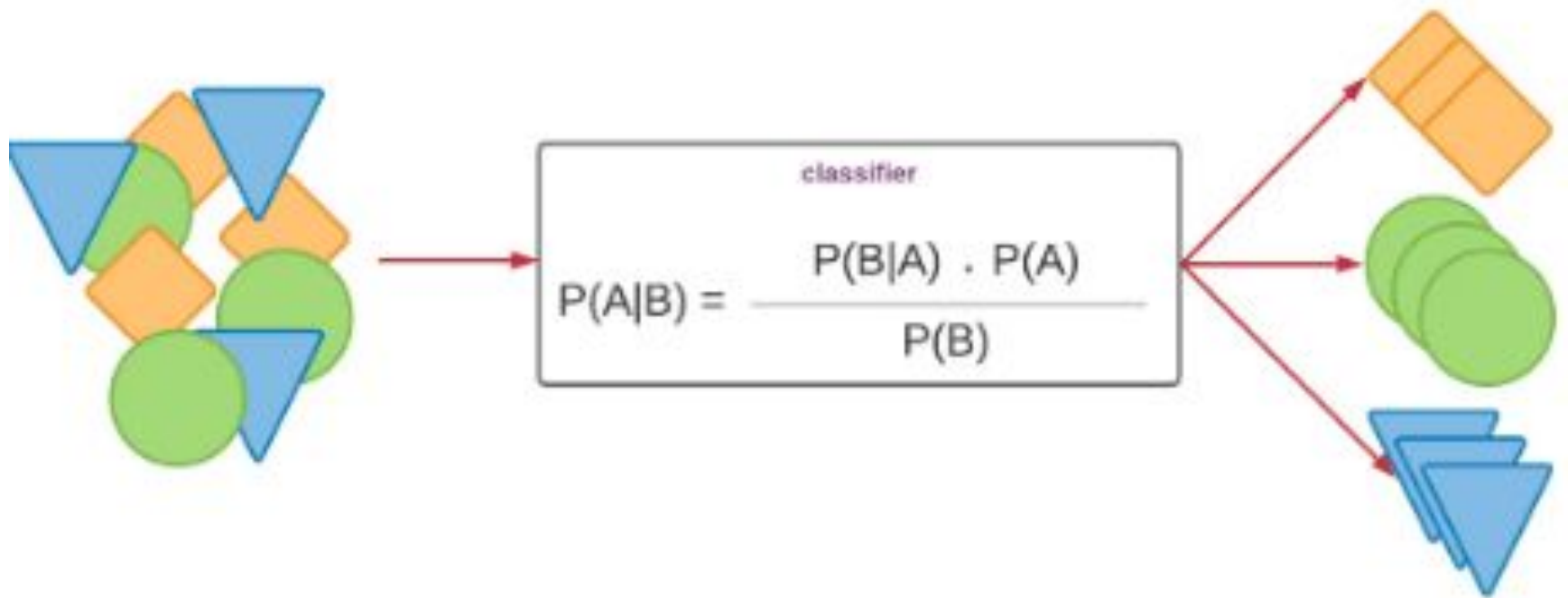
- Probabilistic models use the **idea of probability** to **classify new entities**.
- Probabilistic models see **features and target variables** as random variables.
- The process of modelling represents and manipulates the level of uncertainty with respect to these variables.
- **There are two types of probabilistic models: Predictive and Generative.**
- **Predictive probability models** use the idea of a conditional probability distribution $P(Y | X)$ from which Y can be predicted from X .
- **Generative models** estimate the joint distribution $P(Y, X)$. O

Probabilistic Models

- Once we know the joint distribution for the generative models, we can derive any conditional or marginal distribution involving the same variables.
- Thus, the generative model is capable of creating new data points and their labels, knowing the joint probability distribution.
- The joint distribution looks for a relationship between two variables.
- Once this relationship is inferred, it is possible to infer new data points.
- **Naïve Bayes is an example of a probabilistic classifier**

Naïve Bayes Classifier Algorithm

- Naïve Bayes algorithm is a supervised learning algorithm, which is based on **Bayes theorem** and used for solving classification problems.
- It is mainly **used in *text classification* that includes a high-dimensional training dataset.**
- **It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.**
- Some popular examples of Naïve Bayes Algorithm are **spam filtration, Sentimental analysis, and classifying articles.**
- **Naïve:** It is called Naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features. Such as if the fruit is identified on the bases of color, shape, and taste, then red, spherical, and sweet fruit is recognized as an apple.



Naïve Bayes Classifier Algorithm

- **Bayes:** It is called Bayes because it depends on the principle of Bayes' Theorem.
- Bayes' Theorem: is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.
- The formula for Bayes' theorem is given as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- **Where,**
- **$P(A|B)$ is Posterior probability:** Probability of hypothesis A on the observed event B.
- **$P(B|A)$ is Likelihood probability:** Probability of the evidence given that the probability of a hypothesis is true.

Naïve Bayes Classifier Algorithm

- **P(A) is Prior Probability:** Probability of hypothesis before observing the evidence.
- **P(B) is Marginal Probability:** Probability of Evidence.
- Suppose we have a dataset of **weather conditions** and corresponding target variable "**Play**". So using this dataset we need to decide that whether we should play or not on a particular day according to the weather conditions. So to solve this problem, we need to follow the below steps:
 - Convert the given dataset into frequency tables.
 - Generate Likelihood table by finding the probabilities of given features.

Naïve Bayes Classifier Algorithm

- Now, use Bayes theorem to calculate the posterior probability.
- **Problem:** If the weather is sunny, then the Player should play or not?
- **Solution:** To solve this, first consider the below dataset:

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Naïve Bayes Classifier Algorithm

Frequency table for the Weather Conditions:

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	5

Likelihood table weather condition:

Weather	No	Yes	
Overcast	0	5	$5/14 = 0.35$
Rainy	2	2	$4/14 = 0.29$
Sunny	2	3	$5/14 = 0.35$
All	$4/14 = 0.29$	$10/14 = 0.71$	

Naïve Bayes Classifier Algorithm

- **Applying Bayes'theorem:**
- **$P(\text{Yes} | \text{Sunny}) = P(\text{Sunny} | \text{Yes}) * P(\text{Yes}) / P(\text{Sunny})$**
- $P(\text{Sunny} | \text{Yes}) = 3/10 = 0.3$
- $P(\text{Sunny}) = 0.35$
- $P(\text{Yes}) = 0.71$
- So $P(\text{Yes} | \text{Sunny}) = 0.3 * 0.71 / 0.35 = \mathbf{0.60}$
- **$P(\text{No} | \text{Sunny}) = P(\text{Sunny} | \text{No}) * P(\text{No}) / P(\text{Sunny})$**
- $P(\text{Sunny} | \text{NO}) = 2/4 = 0.5$
- $P(\text{No}) = 0.29$
- $P(\text{Sunny}) = 0.35$
- So $P(\text{No} | \text{Sunny}) = 0.5 * 0.29 / 0.35 = \mathbf{0.41}$
- So as we can see from the above calculation that **$P(\text{Yes} | \text{Sunny}) > P(\text{No} | \text{Sunny})$**
- **Hence on a Sunny day, Player can play the game.**

Advantages of Naïve Bayes Classifier:

- Naïve Bayes is one of the **fast and easy ML algorithms to predict a class of datasets.**
- It can be used for **Binary** as well as **Multi-class Classifications.**
- It performs well in Multi-class predictions as compared to the other Algorithms.
- It is the most popular choice for **text classification problems.**

Disadvantages of Naïve Bayes Classifier:

- Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between features.

Applications of Naïve Bayes Classifier:

- It is used for **Credit Scoring.**
- It is used in **medical data classification.**
- It can be used in **real-time predictions** because Naïve Bayes Classifier is an eager learner.
- It is used in Text classification such as **Spam filtering** and **Sentiment analysis.**

Grouping and Grading Model

- Grading vs grouping is an orthogonal categorization to geometric-probabilistic-logical-compositional.
- Grouping models break the instance space up into groups or segments and in each segment apply a very simple method (such as majority class).
- E.g. **decision tree, KNN.**
- Grading models form one global model over the instance space.
- E.g. Linear classifiers – Neural networks

Parametric Machine Learning Algorithms

- A learning model that summarizes data with a set of **parameters of fixed size** (independent of the number of training examples) is called a parametric model.
- **No matter how much data you throw at a parametric model**, it won't change its mind about how many parameters it needs.
- The algorithms involve two steps:
- Select a form for the function.
- Learn the coefficients for the function from the training data.
- An easy to understand functional form for the mapping function is a line, as is used in **linear regression**:
- $b_0 + b_1 * x_1 + b_2 * x_2 = 0$

Naïve Bayes Classifier Algorithm

- Where **b0**, **b1** and **b2** are the coefficients of the line that control the intercept and slope, and **x1** and **x2** are two input variables.
- a linear combination of the input variables and as such parametric machine learning algorithms are often also called “*linear machine learning algorithms*”.
- **Some more examples of parametric machine learning algorithms include:**
 - Logistic Regression
 - Linear Discriminant Analysis
 - Perceptron Naive Bayes Simple Neural Networks
- **Parameters for using the normal distribution is as follows:**
 - Mean
 - Standard Deviation

Naïve Bayes Classifier Algorithm

- **Benefits of Parametric Machine Learning Algorithms:**
- **Simpler:** These methods are easier to understand and interpret results.
- **Speed:** Parametric models are very fast to learn from data.
- **Less Data:** They do not require as much training data and can work well even if the fit to the data is not perfect.
- **Limitations of Parametric Machine Learning Algorithms:**
- **Constrained:** By choosing a functional form these methods are highly constrained to the specified form.
- **Limited Complexity:** The methods are more suited to simpler problems.
- **Poor Fit:** In practice the methods are unlikely to match the underlying mapping function.

Nonparametric Machine Learning Algorithms

- Nonparametric methods are good when you have a lot of data and no prior knowledge, and when you don't want to worry too much about choosing just the right features.
- Nonparametric methods seek to best fit the training data in constructing the mapping function, whilst maintaining some ability to generalize to unseen data. As such, they are able to fit a large number of functional forms.
- An easy to understand nonparametric model is the k-nearest neighbors algorithm that makes predictions based on the k most similar training patterns for a new data instance.

Nonparametric Machine Learning Algorithms

- Some more examples of popular nonparametric machine learning algorithms are:
- k-Nearest Neighbors Decision Trees like CART and C4.5
- Support Vector Machines
- **Benefits of Nonparametric Machine Learning Algorithms:**
- **Flexibility:** Capable of fitting a large number of functional forms.
- **Power:** No assumptions (or weak assumptions) about the underlying function.
- **Performance:** Can result in higher performance models for prediction.
- **Limitations of Nonparametric Machine Learning Algorithms:**
- **More data:** Require a lot more training data to estimate the mapping function.
- **Slower:** A lot slower to train as they often have far more parameters to train.
- **Overfitting:** More of a risk to overfit the training data and it is harder to explain why specific predictions are made.

Parametric model	Non-parametric model
Constant number of parameters, independent of training data	Number of parameters grows with the number of training samples
Strong assumption about the training data	No assumption about the training data
Fewer training samples required	Many training samples required
Fast training, fast inference	Slow training and slow inference
Examples: Linear regression and logistic regression	Examples: Decision trees and k-nearest neighbors

Parametric Methods

Parametric Methods uses a fixed number of parameters to build the model.

Parametric analysis is to test group means.

It is applicable only for variables.

It always considers strong assumptions about data.

Parametric Methods require lesser data than Non-Parametric Methods.

Parametric methods assumed to be a normal distribution.

Non-Parametric Methods

Non-Parametric Methods use the flexible number of parameters to build the model.

A non-parametric analysis is to test medians.

It is applicable for both – Variable and Attribute.

It generally fewer assumptions about data.

Non-Parametric Methods requires much more data than Parametric Methods.

There is no assumed distribution in non-parametric methods.

Parametric data handles – Intervals data or ratio data.

But non-parametric methods handle original data.

Here when we use parametric methods then the result or outputs generated can be easily affected by outliers.

When we use non-parametric methods then the result or outputs generated cannot be seriously affected by outliers.

Parametric Methods can perform well in many situations but its performance is at peak (top) when the spread of each group is different.

Similarly, Non-Parametric Methods can perform well in many situations but its performance is at peak (top) when the spread of each group is the same.

Parametric methods have more statistical power than Non-Parametric methods.

Non-parametric methods have less statistical power than Parametric methods.

As far as the computation is considered these methods are computationally faster than the Non-Parametric methods.

As far as the computation is considered these methods are computationally slower than the Parametric methods.

Examples: Logistic Regression, Naïve Bayes Model, etc.

Examples: KNN, Decision Tree Model, etc.

Important Elements in Machine Learning

- **Data formats**
- In a supervised learning problem, there will always be a dataset, defined as a **finite set of real vectors with m features each**:

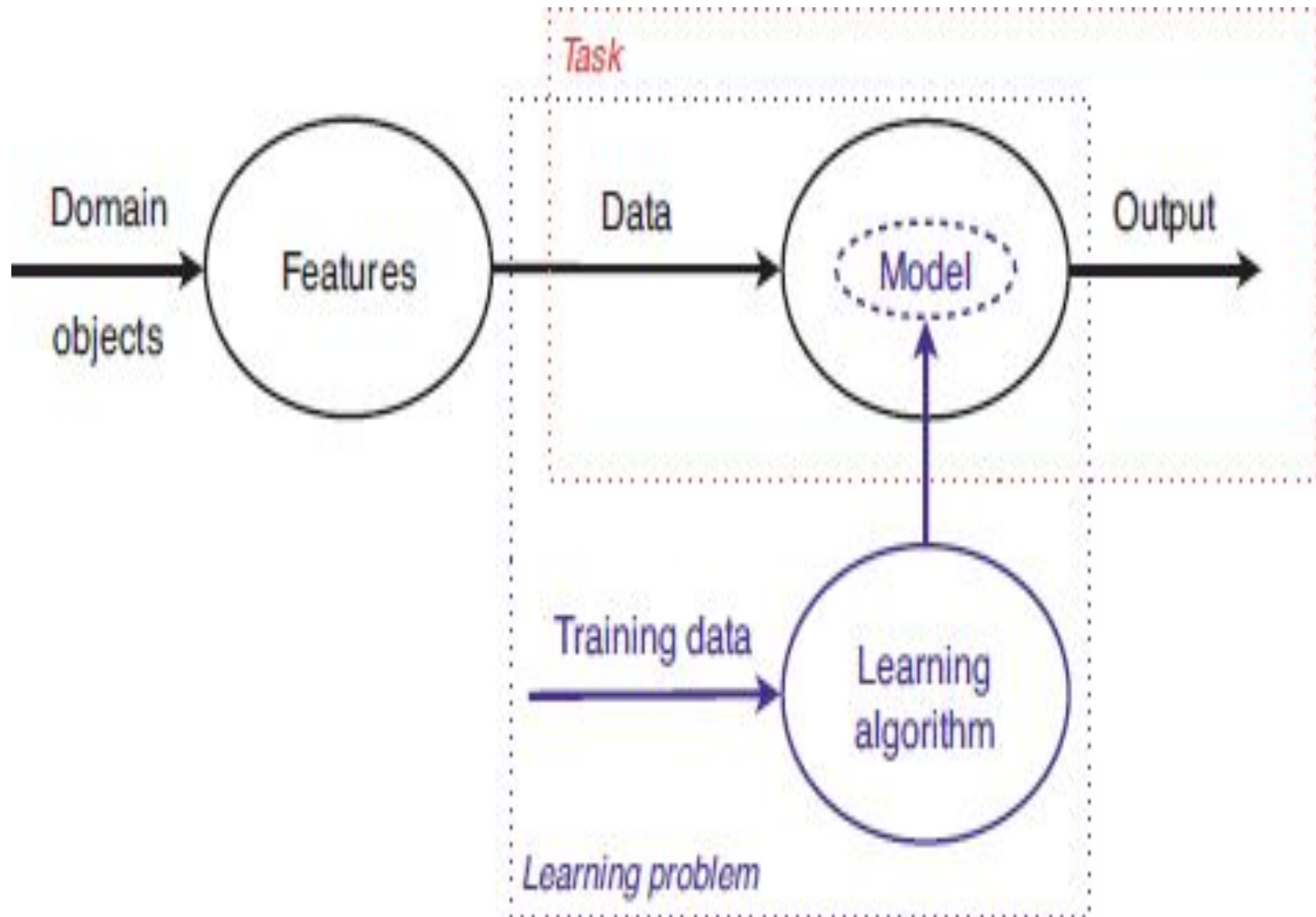
$$X = \{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n\} \text{ where } \bar{x}_i \in \mathbb{R}^m$$

Data Format

- **Labeled data:** Data consisting of a set of *training examples*, where each example is a *pair* consisting of an **input and a desired output value** (also called the *supervisory signal, labels, etc*)
- **Classification:** The goal is to predict discrete values, e.g. $\{1,0\}$, $\{\text{True}, \text{False}\}$, $\{\text{spam}, \text{not spam}\}$.
- **Regression:** The goal is to **predict continuous values**, e.g. home prices.

- **Feature vector:** A typical setting for machine learning is to be given a **collection of objects** (or data points), each of which is **characterised by several different features**.
- Features can be of different sorts: e.g., they might be **continuous** (say, **real- or integer-valued**) or **categorical** (for instance, a feature for **colour** can have values like **green, blue, red**).
- A vector **containing all of the feature values for a given data point** is called the **feature vector**;
- if this is a vector of **length m** , then one can think of each data point as being mapped to a **m -dimensional vector space** (in the case of real-valued features, this is $\mathbb{R}^m$), called the **feature space**.

The elements of Machine Learning



ML- Data Formats

- In machine learning, data formats refer to the way data is organized and represented for training and inference.
- The choice of data format depends on the type of machine learning model being used and the specific requirements of the problem at hand.
- Some common data formats used in machine learning
 - Tabular Data
 - Image Data
 - Text Data Format
 - Time Series Data Format
 - Audio Data Format
 - Video/Frame Data Set
 - Graph Dataset

Tabular Data Formats

- This format is commonly used for **structured data**, where each row represents a single data point, and each column represents a specific feature or attribute.
- Tabular data is typically stored in formats like **CSV (Comma Separated Values)** or **TSV (Tab Separated Values)**.
- Example: A dataset of housing prices, where each row contains information about a house (e.g., number of rooms, location, square footage) and the last column contains the target variable (price).

Image Data Formats

- Images are represented as matrices of pixel values. Depending on the **color channels**, **images can be grayscale (1 channel) or color images (3 channels - RGB).**
- Common formats include **JPEG, PNG, or even raw pixel values as arrays.**
- Example: A dataset of cat and dog images, where each image is stored in JPEG format and labeled as either cat or dog.

Text Data Formats

- Text data can be represented as plain text, tokenized sequences, or numerical vectors using techniques like **word embeddings** or **TF-IDF (Term Frequency-Inverse Document Frequency)**.
- Example: A collection of customer reviews for a product, where each review is a string of text, or tokenized into a sequence of words.

Time Series Data Formats

- Time series data represents **data points ordered by time**.
- Each data point typically contains a **timestamp and one or more feature values**.
- Example: Stock market data with timestamps and corresponding stock prices for different companies.

Audio Data Formats

- Audio data can be represented as **waveforms or spectrograms**.
- Waveforms represent the amplitude of the **audio signal over time**, while **spectrograms represent the frequency components of the audio over time**.
- Example: Speech recognition, where audio clips of spoken words are represented as spectrograms.

Video Data Formats

- Video datasets consist of **sequences of frames (images) captured over time.**
- Each frame represents a **single data point**, and the entire video dataset is composed of multiple sequences, where **each sequence captures a specific event or action.**
- **Example:** A dataset of human activities, such as walking, running, eating, or playing sports, where each video captures a person performing a specific action.

Graph Data Formats

- Graph data represents **entities and their relationships in a network.**
- Nodes represent entities, and **edges represent relationships between nodes.**
- Example: Social network data, where users are nodes, and connections between users are edges.

Learnability in Machine Learning

- **Example A: 2, 4, 6, 8**
- **Predict Next Number**
- **the rule is “+2”**
- **It is obvious and easy to predict**

Learnability in Machine Learning

- **Example A: 6,5,1,3**
- **Predict Next Number**
- **no obvious rule or pattern.**
- **It is obvious and easy to predict**

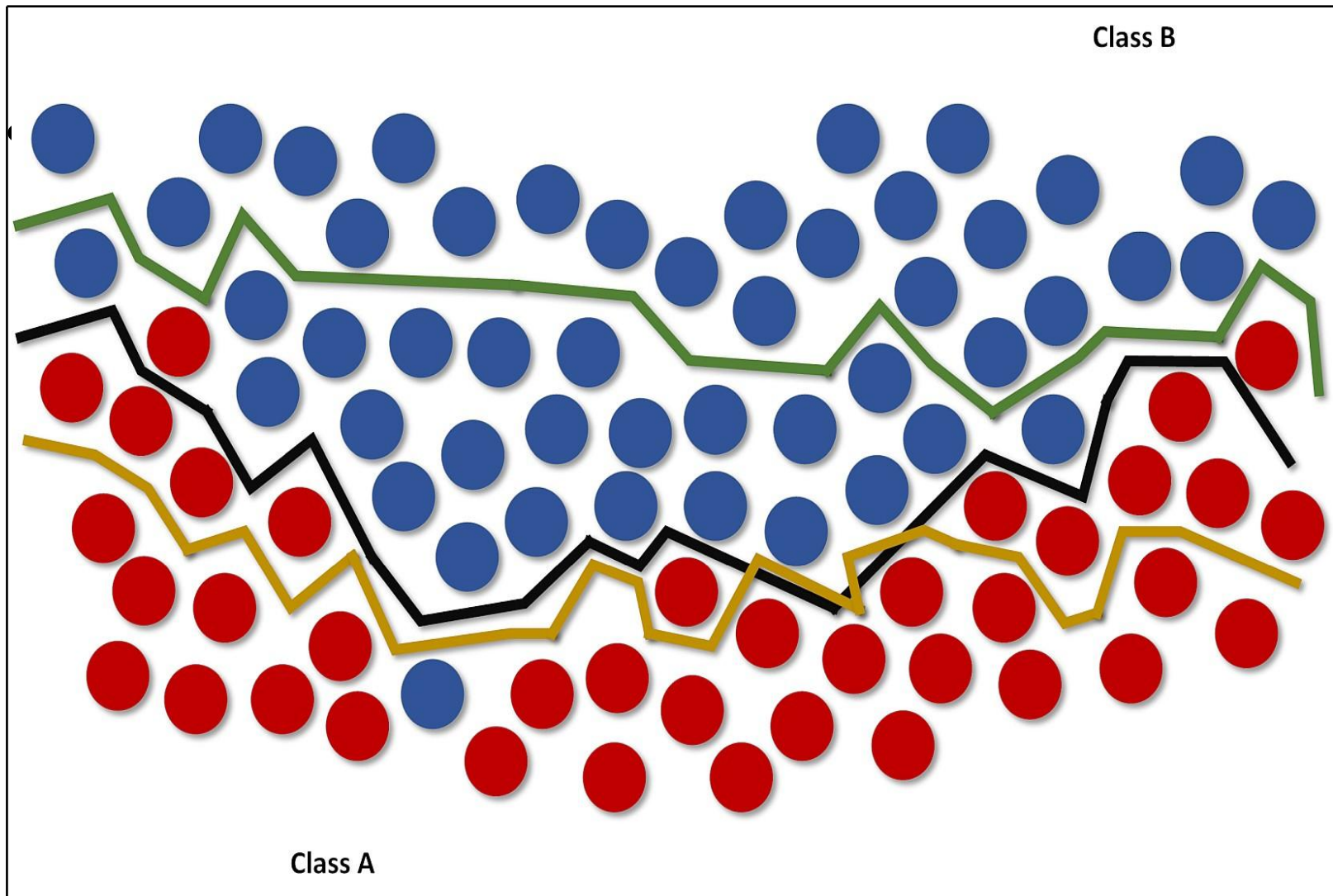
Learnability in Machine Learning

- Learnability in machine learning **refers to the ability of an algorithm or model to learn and generalize from the training data to make accurate predictions on new, unseen data.**
- In other words, a learnable machine learning algorithm should be able to **extract patterns and relationships from the training data** and apply that knowledge to make predictions on previously unseen data points.

Learnability in Machine Learning

- The ultimate goal of any machine learning task is to **develop models that can perform well on real-world data**, not just on the data used for training.
- This is achieved through the **process of generalization**, where a model can adapt to new data that it has never encountered before. A model that can generalize well is said to be "learnable."

Learnability



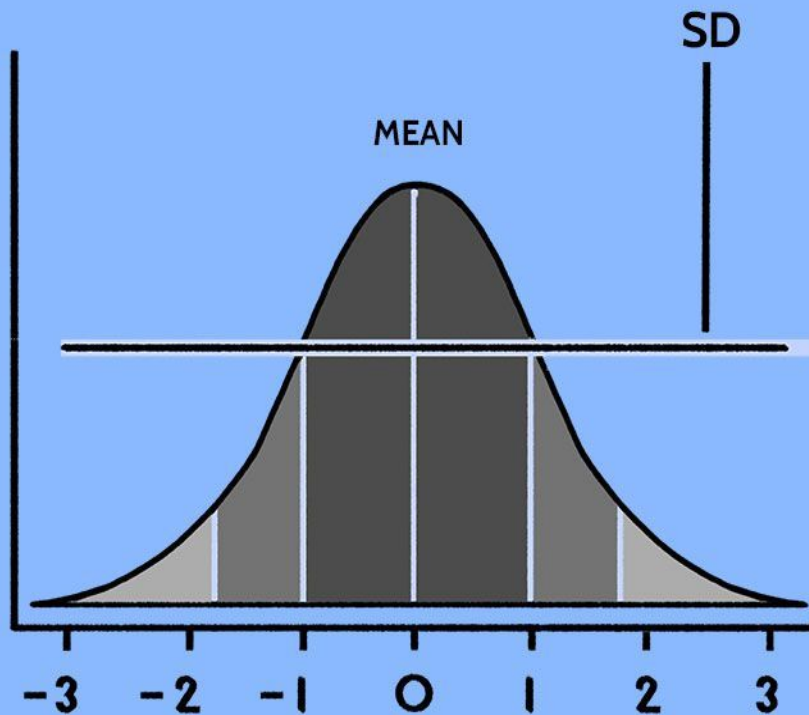
- there's an example of a dataset whose points must be classified as red (**Class A**) or blue (**Class B**).
- Three hypotheses are shown: the first one (the middle line starting from left) misclassifies one sample,
- while the lower and upper ones misclassify **13** and **23** samples respectively:
- the first hypothesis is optimal and should be selected; however, it's important to understand an essential concept which can determine a potential overfitting

Statistical learning approaches

- The statistics for machine learning come along in the **first phase of machine learning algorithm.**
- It helps us **to deal with the data**, as it is the foundation of implementing statistical concepts and ultimately interpreting conclusions from it.
- To simplify, **it is a mathematics field used for collecting, organizing, and analyzing data.**
- its subcategories are:
 - Descriptive statistics
 - Inferential statistics

Statistical learning approaches

- **Descriptive statistics:**
 - **used to summarize and describe the main features of a dataset.**
 - In the context of machine learning, descriptive statistics help to **gain insights into the characteristics of the data before building and training a model.**
 - Some common measures used in descriptive statistics include: Min, Median, Standard Deviation, **(A standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean. Low standard deviation means data are clustered around the mean, and high standard deviation indicates data are more spread out.)** Percentage etc.



Standard Deviation

[ˈstan-dərd dē-vē-ˈā-shən]

A statistic that measures the dispersion of a dataset relative to its mean and is calculated as the square root of the variance.

Statistical learning approaches

- **Inferential statistics**
 - used to **draw inferences or make predictions about a population based on a sample of data.**
 - In machine learning, inferential statistics are often employed to **make predictions on new, unseen data using a model trained on a representative sample.**
 - The process of inferential statistics involves the following steps
 - Random sampling
 - Model Building
 - Model Evaluation

Statistical learning approaches

- **Inferential statistics**

- used to **draw inferences or make predictions about a population based on a sample of data.**
- In machine learning, inferential statistics are often employed to **make predictions on new, unseen data using a model trained on a representative sample.**
- The process of inferential statistics involves the following steps
 - Random sampling
 - Model Building
 - Model Evaluation

How Statistics is Used in Machine Learning

